

Vertically Conducting β -Ga₂O₃ Diodes on 4H-SiC Substrate as Short Wavelength (<254 nm) Switches/Detectors

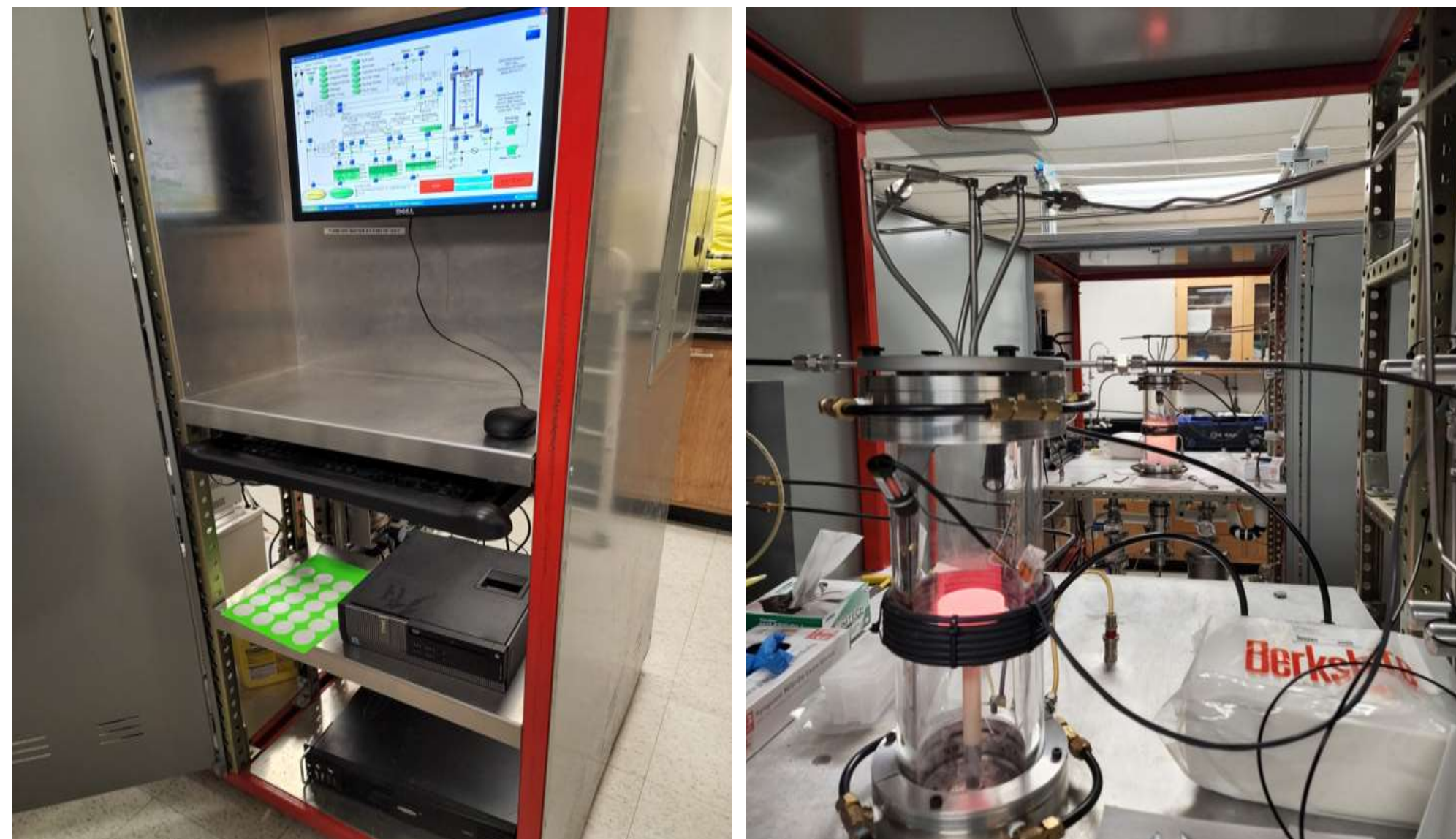
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Introduction

- Vertically conducting diodes were designed and fabricated using metal organic chemical deposition (MOCVD) grown layers of β -Ga₂O₃ on n-4H-SiC substrate.
- Co-delta doping with Si and In is used to improve crystallinity.
- In & Si co-delta doped samples were analyzed using deep-UV 254nm radiation - above the bandgap of the grown material.
- The temperature-dependent behavior of the samples studied.
- The designed diode structures show remarkable responses to 254 nm wavelength, making them excellent solar blind detectors.

MOCVD Systems

- The lab at the **University of South Carolina** used a dedicated MOCVD system for III-oxide growth and is capable of growing:
 - **Compound of III-oxides:** such as (Al_xIn_yGa_{1-x-y})₂O₃ with n-type doping (Si), both in continuous and pulsed modes.
 - **MOCVD III-Oxides System:** Can grow III-oxides on substrates like sapphire, Si, and SiC at a growth rate of 0.5 to 4 μ m/hr and can create different device structures for power electronics and radiation detection.



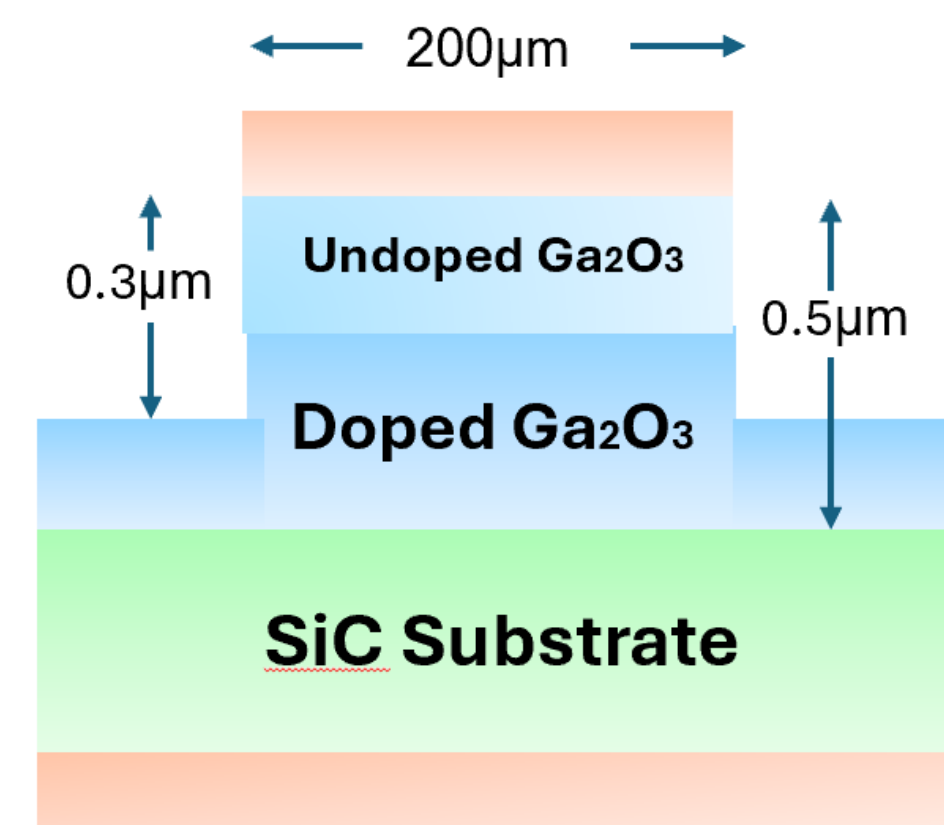
- Moreover, the lab is equipped with all necessary characterization techniques, including UV-vis spectroscopy, Hall measurements, C-V profiling, AFM, and temperature-dependent electrical analysis, for comprehensive material and device evaluation.

Materials & Growth Parameters

- **Substrate Selection:** A 4° miscut n-type 4H-SiC substrate was used, which has higher thermal conductivity than β -Ga₂O₃, and the high thermal conductivity of the substrate is expected to aid heat dissipation in vertical device structures.
- **Precursor Chemistry:** Triethyl gallium (TEGa) served as the gallium precursor, ultra-high purity O₂ ensured stoichiometric oxide formation, silane (SiH₄) (10 ppm diluted with nitrogen) provided controlled n-type doping, and trimethyl indium (TMIn) was used as an In source that expected to act as a surfactant to improve adatom mobility and reduce defect formation.
- **Optimized Growth Conditions:** The MOCVD growth was carried out at 680°C and 50Torr, conditions carefully chosen to balance precursor decomposition, reduce parasitic gas-phase reactions, and yield smooth, high-quality films. The growth rate of the layers was 0.75 μ m/hr. A closed-loop feedback controlled the growth temperature, confirmed by a disappearing-filament pyrometer.

Device Structure

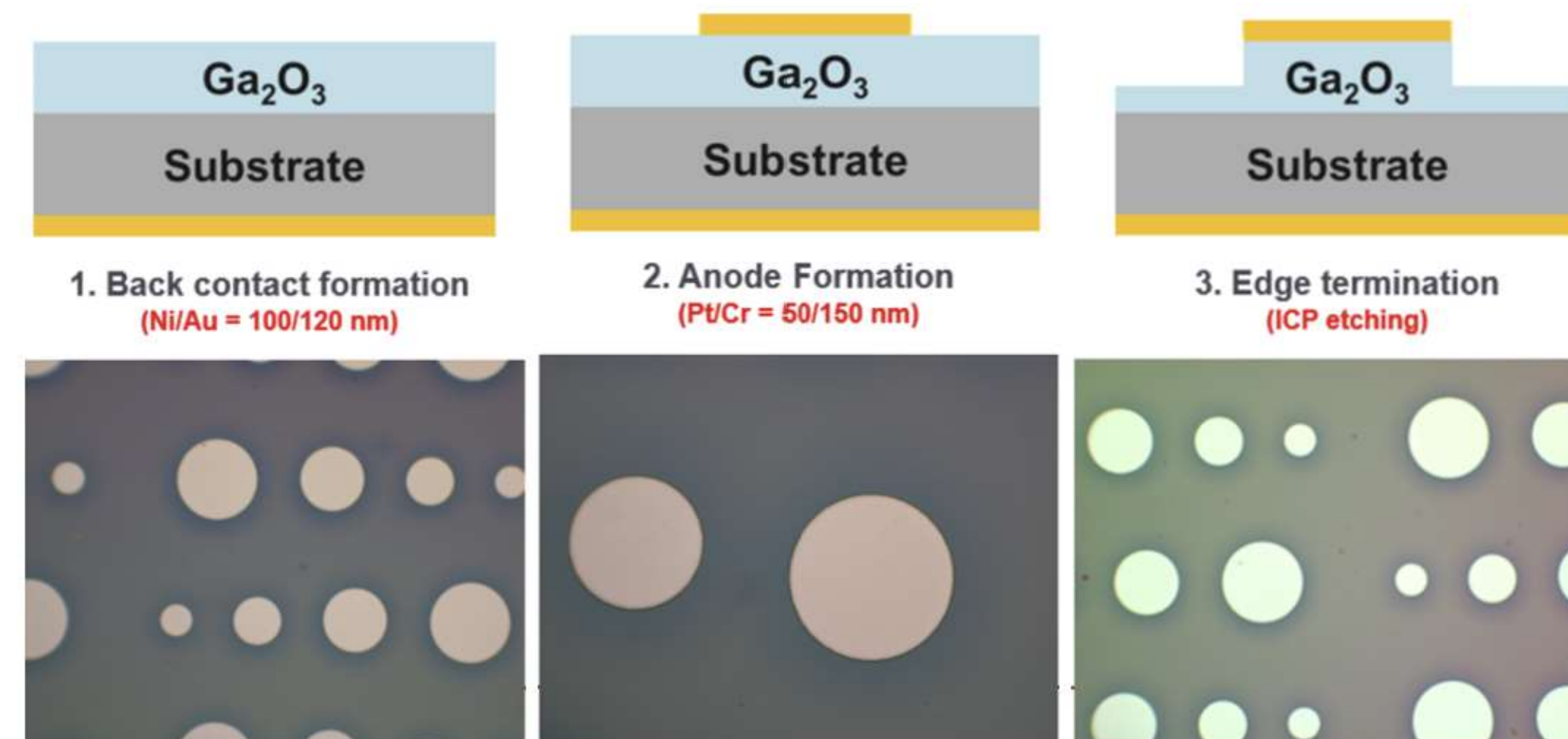
- **Substrate and Epitaxial Layers:** The device is fabricated on a 4° off-axis 4H-SiC substrate at the **University of Utah (UoU)** to enhance thermal dissipation. The total thickness of the device structure is 0.5 μ m.
 - 0.25 μ m doped β -Ga₂O₃ on a conducting substrate was grown, and after that, a 0.25 μ m undoped layer was grown.
 - The scheme was adopted to give a low current when it is not irradiated by the above bandgap radiation.
- **Doping Strategy:** The bottom β -Ga₂O₃ layer is co-delta doped with silicon (Si) and indium (In) using a repeated pulsed doping cycle—0.1min SiH₄ + TMIn followed by 0.2min TMGa + O₂—to enhance crystalline quality and electron mobility.
- **Functional Design:** The undoped top layer acted as a pseudo p-type with high resistivity and suppresses current, while the doped bottom layer creates a diode structure. This heterostructure supports high-selectivity UV-C detection by balancing electrical isolation and conductivity



Device Structural Fabrication

The device layers were processed to fabricate the diode structure at UoU.

- **Back Contact Formation:** Ni/Au (100/120nm) is deposited on the backside via e-beam evaporation to form the ohmic contact. The Ohmic check using the transmission line measurement (TLM).
- **Top Contact & Patterning:** Pt/Cr (50/150nm) is deposited on the top surface using a lift-off process to define circular anodes. The diameters of the diodes were varied – we only fully analyzed 200 μ m diode for this study.
- **Mesa Etch Isolation:** ICP etching (~300nm) creates device mesas for edge termination, reducing leakage and preventing premature breakdown.



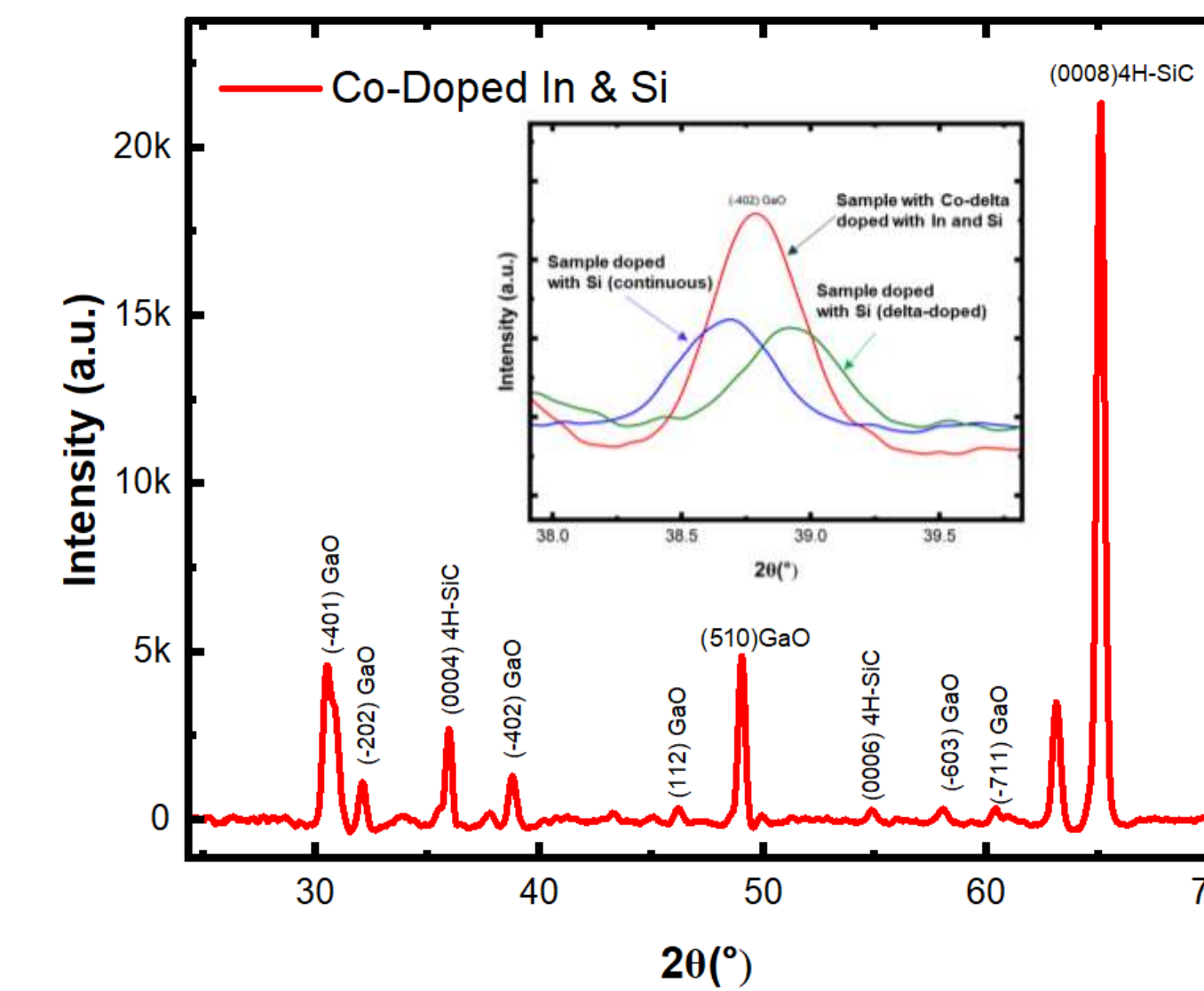
Fabrication process of the diodes at the University of Utah.

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Result and Discussion

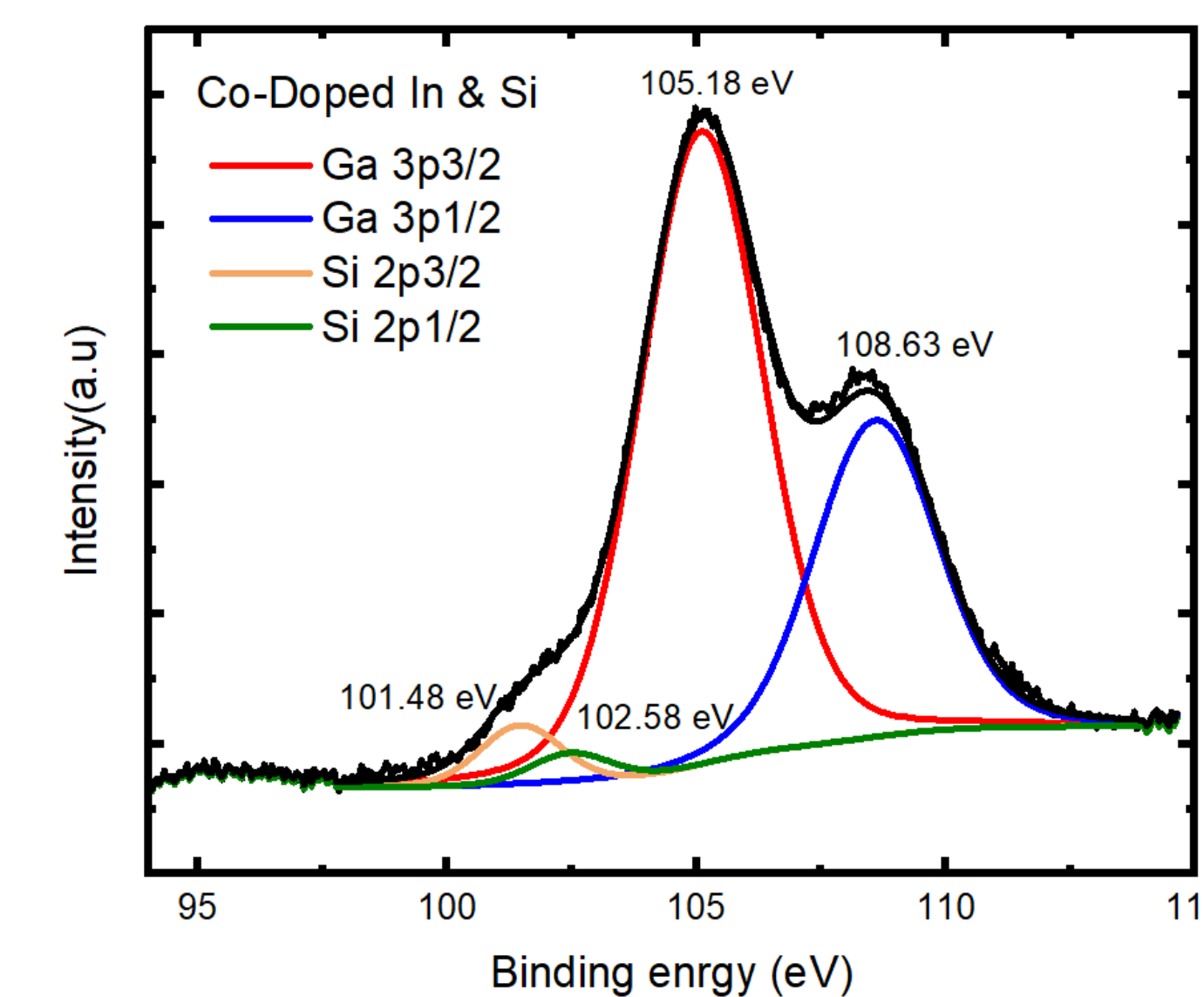
XRD Analysis

- **Substrate and Phase Identification:** XRD confirms the formation of monoclinic β -Ga₂O₃ with prominent diffraction peaks corresponding to the (-402) and (-603) planes, indicating preferential crystalline orientations.
- **Mixed-Phase Detection:** Minor ϵ -Ga₂O₃ peaks observed around 32° and 60° suggest the presence of a secondary phase due to mixed-phase formation at MOCVD growth temperatures.
- **Substrate Peaks:** Distinct peaks at 35.9°, 41.5°, and 65.9° match the (0004), (0006), and (0008) reflections of 4H-SiC, confirming successful heteroepitaxial growth on the SiC substrate.
- **Crystallinity Improvement:** A narrow full width at half maximum (FWHM) of 0.44° for the (-402) peak indicates reduced defect density and enhanced crystallinity—attributed to the co-delta doping of Si and In.



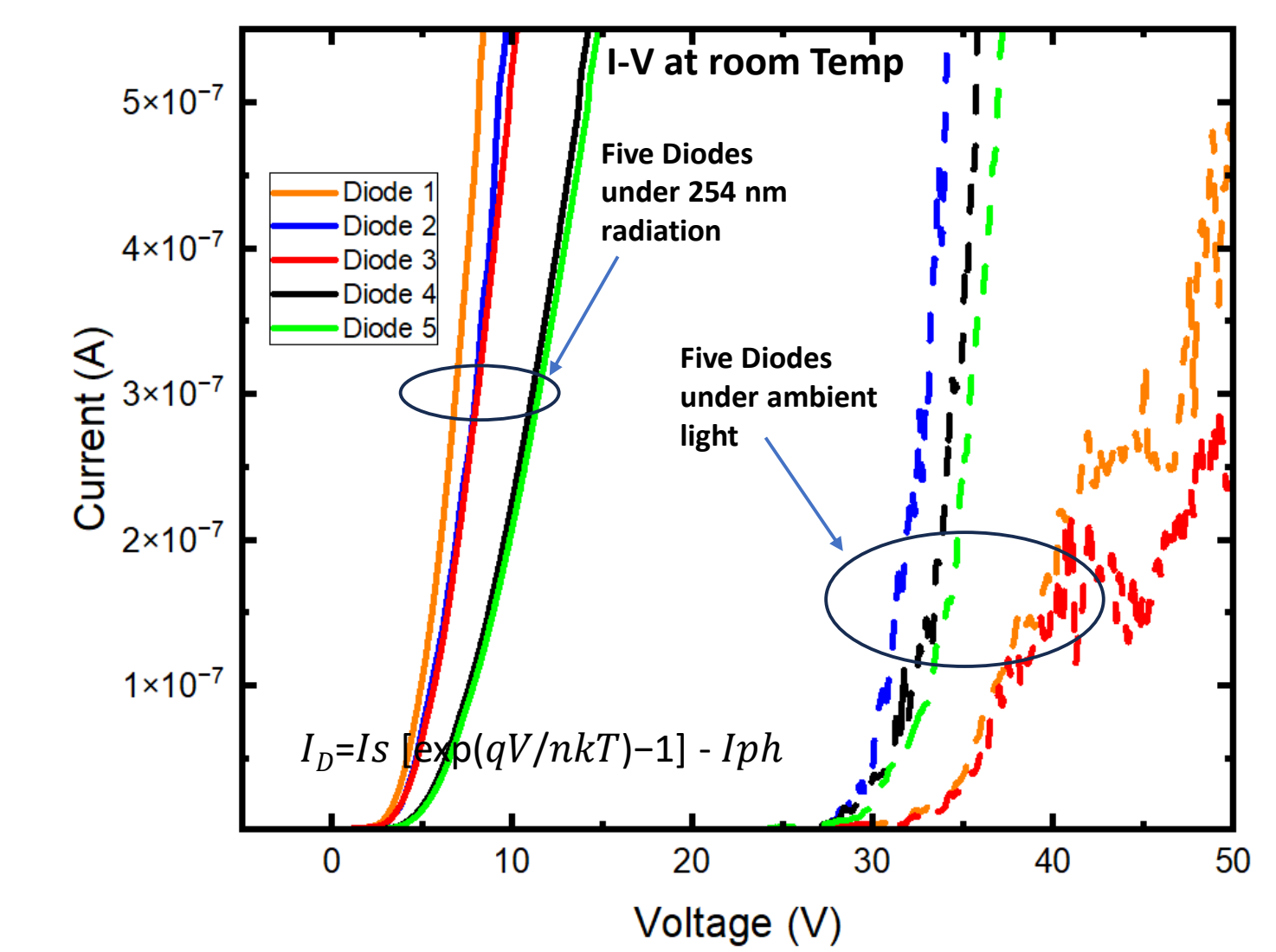
XPS Analysis

- **Elemental Confirmation:** Ga 3p peaks at 105.18eV and 108.63eV confirm the Ga³⁺ oxidation state, while Si 2p peaks at 101.48eV and 102.58eV validate successful silicon incorporation consistent with β -Ga₂O₃ and SiO₂ environments.
- **Surface Stability & Doping Integrity:** No significant peak shifts were observed, indicating chemical stability and absence of phase change; weak In 3d signals (<1%) confirm trace indium incorporation, supporting successful co-doping without structural disruption.



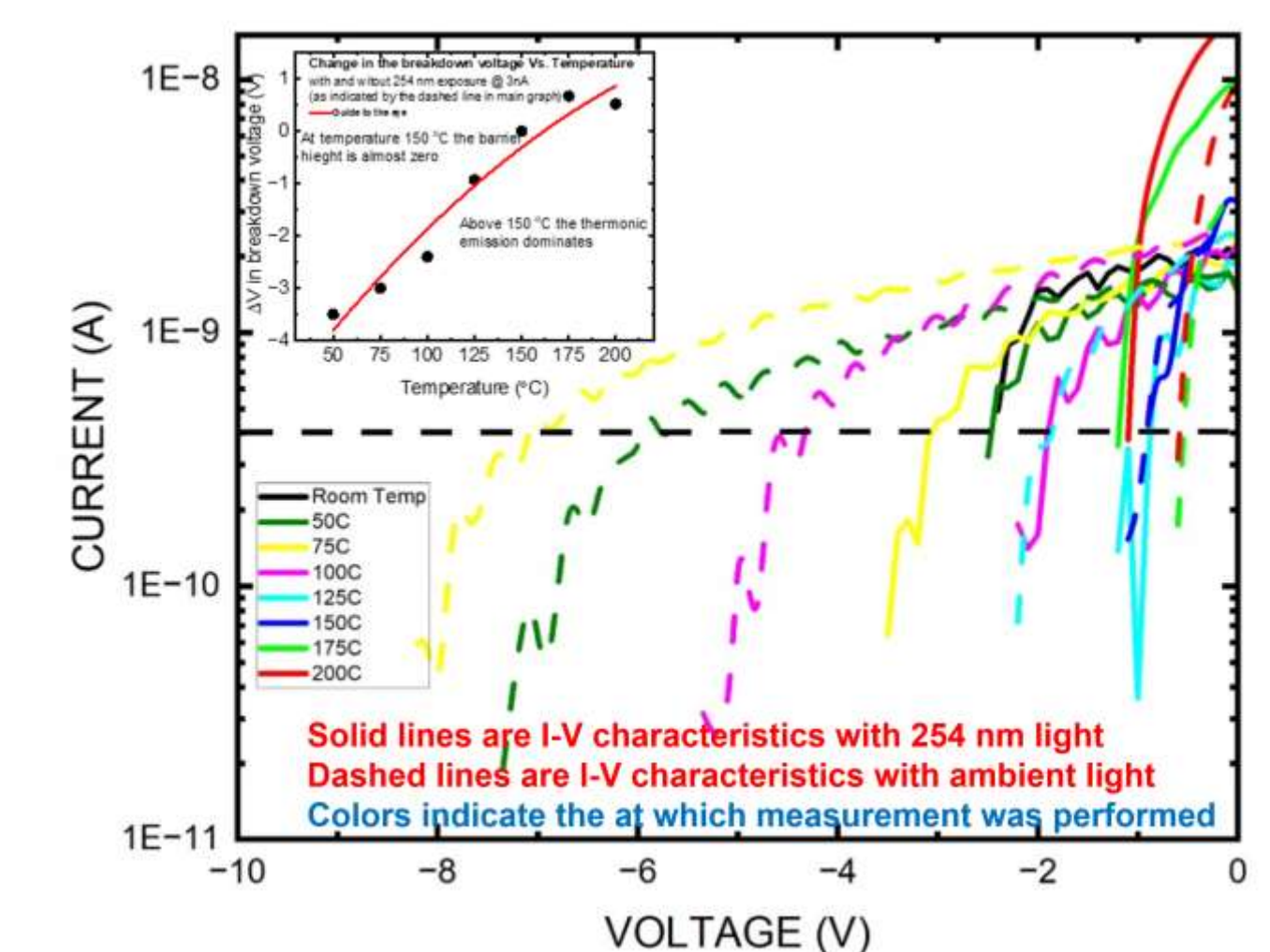
Solar Blind Response

- Substrate Under 254nm illumination, the devices exhibit a 10²–10³× increase in photocurrent compared to ambient conditions.
- The diode biased at 10 V can be switched on with 254nm light, only providing a solar-blind switch.
- UV-generated carriers reduce the turn-on voltage from ~35V to ~2 to 5V, indicating barrier height lowering – allude to switching behavior. All five co-delta-doped diodes exhibit a consistent response, validating the process uniformity and reproducibility.



Thermal Study of the Diodes

- As temperature increases, reverse breakdown voltage occurs at the lower voltage due to enhanced thermionic emission, while reverse leakage escalates above 125°C from trap-assisted conduction and dislocation activation at the Ga₂O₃/SiC interface.
- Devices maintain stable diode characteristics up to ~150°C, but degradation in junction isolation and non-ideal I–V behavior emerges beyond 150 °C, indicating dominant thermionic emission.



Conclusion

- Forward-biased Ga₂O₃ diodes are designed to act as a light-activated switch that can only be activated by wavelengths shorter than 254 nm.
- The diodes behave properly up to temperatures of around 150 °C; above 150 °C, thermionic emission dominates.
- Thus, these devices are applicable for switching optically large currents with proper heat removal through the SiC substrate.

Note: References could not be provided due to space limitations.



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